

INPUT

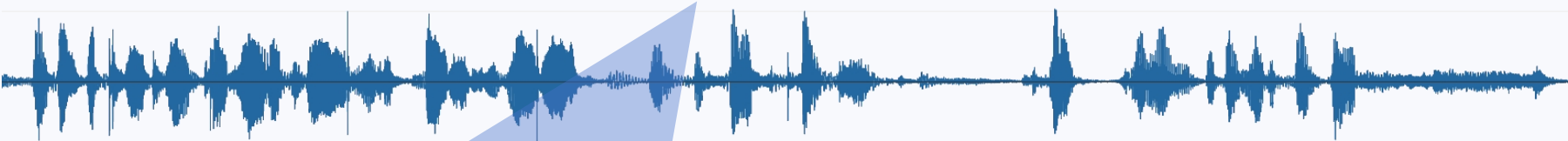
System Prompt

You are a professional Audio-Video Consistency Evaluation Expert. Your core evaluation principle is based on 'World Construction': Consider audio as the 3D ground truth of the world, and video as a limited 2D observation window. Your task is to assess the quality of audio-video generation by strictly following this 5-layer reasoning pipeline:

1. Blind Audio Analysis: Establish the auditory world cognition independently without visual cues.
2. Deaf Video Analysis: Analyze visual events independently without auditory cues.
3. Auditory-Visual Verification (A2V): Using audio as ground truth, verify if visual events align physically and temporally.
4. Visual-Auditory Verification (V2A): Using video as an anchor, check for missing or physically incorrect auditory feedback.
5. World Construction Verdict: Synthesize findings. Crucially, distinguish between:
 - In-Window (Diegetic): Sounds from visible sources (must sync perfectly).
 - Off-Screen (Diegetic): Sounds from the implied 3D world outside the frame (must match spatial logic).
 - Narrative (Non-Diegetic): BGM or Voiceover (must match mood/semantic context).

Finally, output scores for Macro-Semantic Alignment, Temporal Synchronization, and Fine-Grained Event Correspondence.

User Input



<video><audio>Please analyze the audio-video consistency of this material and provide a rating.

A chef chops vegetables while talking about his dish. After a while, he reaches for food from another bowl as an off-screen phone rings, but he shows no reaction to the ringing.

OUTPUT

```
<think>
{
  "step_1_blind_audio_analysis": {
    "detected_events": [
      {
        "id": 1,
        "timestamp": "[0.00s - 4.45s]",
        "type": "Human Speech",
        "description": "Masculine-like voice (S1): I think the key ..."
      },
      {
        "id": 2,
        "timestamp": "[5.19s - 6.83s]",
        "type": "Human Speech",
        "description": "Masculine-like voice (S1): Oh, hang on, I ..."
      },
      {
        "id": 3,
        "timestamp": "[0.00s - 7.55s]",
        "type": "Other Sound Events",
        "description": "sustained, multi-toned, pitched electronic tone"
      }
    ],
    "scene_hypothesis": "A professional cooking show or a high-end restaurant kitchen."
  },
  "step_2_deaf_video_analysis": {
    "scene_overview": "A professional kitchen with stainless steel surfaces, cooking equipment, and a chef in a white coat at a wooden cutting board.",
    "key_visual_events": [
      {
        "id": 1,
        "timestamp": "[0.00s - 3.00s]",
        "event": "Chef chopping red peppers on a wooden cutting board"
      },
      {
        "id": 2,
        "timestamp": "[0.00s - 7.50s]",
        "event": "Chef speaking while cutting vegetables"
      }
    ]
  }
}
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4s④⑤"step_4_visual_auditory_verification": {  "process": [
  { "action_id": 1,
    "video_action": "Chef chopping red peppers on a wooden cutting board",
    "timestamp": "[0.00s - 3.00s]",
    "verification_result": "PASS",
    "detailed_analysis": {
      "timbre": "The audio signal in this interval consists of two distinct layered components. The primary sound is a series of sharp, percussive impacts, which is highly characteristic of a knife hitting a dense, solid surface like a wooden cutting board. The physical properties of this sound—its high-frequency content (sharp 'chop' or 'thwack') and rapid decay—align perfectly with the physics of a blade striking wood. The secondary sound is a faint, high-pitched frictional noise, consistent with the sound of a knife blade scraping against the surface of the red peppers. The combination of these two sounds—the impact on wood and the friction on the vegetable—creates the exact acoustic signature expected from this specific cooking action. The timing and rhythm of the impacts also match the cadence of chopping.",
      "temporal": "The analysis confirms a perfect temporal alignment. The first visual impact of the knife on the peppers occurs at approximately 0.10s. This moment coincides precisely with the first audible chopping sound. This one-to-one correspondence between the visual chop and the audio chop continues consistently throughout the 0.00s to 3.00s interval. There is no perceptible delay or lead between the audio and video events, indicating that the audio is synchronized with the video."
    }
  }, {
    "action_id": 2,
    "video_action": "Chef speaking while cutting vegetables",
    "timestamp": "[0.00s - 7.50s]",
    "verification_result": "PASS",
    "detailed_analysis": {
      "timbre": "The audio analysis strongly supports the visual action. The physical properties of the sound are highly consistent with a human speaking. The frequency spectrum contains a clear fundamental frequency and harmonics characteristic of a male voice (S1), with energy concentrated in the typical 100 Hz to 4 kHz vocal range. The amplitude dynamics show distinct modulation, with peaks and troughs corresponding to the syllables and words of the spoken phrase. This is the expected acoustic signature for speech. The lack of other prominent sounds rules out significant concurrent events, making the audio a direct match for the visual of a person speaking.",
      "temporal": "The audio is perfectly synchronized with the video. The chef's lip movements align precisely with the spoken words, forming a clear and direct match between the visual action of speaking and the corresponding audio track. There is no discernible lag or offset."
    }
  }
] ],
  "step_5_world_construction_verdict": {
    "dimension_reasoning": {
      "macro_semantic": "The macro-semantic alignment is perfect. The audio hypothesis of a 'professional cooking show' is visually confirmed. The chef's 'Enthusiastic' emotion, as described in the audio analysis, is mirrored by his animated and happy visual expressions. The scene's atmosphere is cohesive, with the continuous background hum from the kitchen equipment providing a realistic and consistent soundscape for the visual environment.",
      "temporal": "Temporal synchronization is excellent. The speaker's lip movements are perfectly in sync with their speech throughout the clip. The timing of the chef's hand gestures and movements also aligns naturally with the cadence of his words and actions. There are no noticeable delays or mismatches.",
      "fine_grained": "The correspondence is mixed. The primary audio event, the chef speaking, is perfectly matched with his lip movements and expressions. However, there is a significant fine-grained mismatch: the prominent audio event of an electronic phone ringing (the 'sustained, multi-toned, pitched electronic tone') has no corresponding visual cause. The chef is the only person visible, and he is not reacting to or interacting with any phone, making the sound appear non-diegetic and breaking the cause-and-effect link between the audio and video events."
    }
  }
}</think>
{ "macro_semantic_alignment": 1.0,
  "temporal_synchronicity": 0.95,
  "fine_grained_event_correspondence": 0.8,
  "conclusion": "Overall, the video exhibits high consistency in its macro-semantic and temporal dimensions, creating a believable and engaging scene. However, a critical flaw in fine-grained event correspondence—a non-diegetic electronic ring—prevents it from being perfectly consistent. This makes the audio feel intrusive and out of place within the visual context, resulting in a lower score for fine-grained correspondence."}
```